

CURRICULUM VITAE

Sagar J C

Contact Details: +91 7892256415 | jcsagar195@gmail.com | sagar.jc@tifr.res.in

Web Page: <https://sagarjc.github.io/>

EDUCATION

- **M.Sc. Physics** August 2022 - May 2024
National Institute of Technology Karnataka
CGPA: 10/10
Received "Board Chairman's Gold Medal" for scoring the highest CGPA 10/10 in the graduating batch of 2023-24. Received the "K. Subbarayappa Gold Medal" for having the best academic record in M.Sc. Physics. Surathkal, India
- **B.Sc. in Physics, Chemistry, and Mathematics** June 2019 - May 2022
St. Joseph's College
Percentage: 97.86%
Received Gold Medals in Physics and Chemistry, and a Silver Medal in Mathematics in the batch of 2019-22. Bengaluru, India
- **Pre-University Course in Physics, Chemistry, Mathematics, and Biology** June 2017 - March 2022
St. Joseph's Pre-University College
Percentage: 96.67%
Awarded Subject Topper in Physics and Chemistry, and ranked second in Mathematics in the batch of 2017-19. Bengaluru, India

PUBLICATIONS

- [1] Sagar J C, Karthik R, Kartheek Hegde, et al., *Perturbations of Black Holes Surrounded by Anisotropic Matter Field*, In: Phys. Rev. D, Vol: 111 (6 Mar. 2025), p. 064034., DOI: <https://doi.org/10.1103/PhysRevD.111.064034>

EXPERIENCE

- **Tata Institute of Fundamental Research, Hyderabad.** August 2024 - July 2025
Research Scholar in the Subject Board of Physics
Hyderabad, India
 - Completed first-year Ph.D. coursework with an aggregate CGPA of 9.1/10, earning 54 credits.
 - Conducted theoretical research on Quantum Spin Liquids (condensed matter physics), under the supervision of Prof. Darshan G. Joshi (Jan 2025 – May 2025).
 - Served as Teaching Assistant for Quantum Mechanics I (Jan 2025 – Apr 2025), conducting weekly tutorial sessions, assisting students in problem-solving. Evaluating and grading their assignments.
- **Quantum Materials Lab, CeNSE, Indian Institute of Science.** May 2023 - July 2023
Research Internship as a part of Q-Daksha Program
Bengaluru, India
 - Completed a Research Internship offered by Quantum Research Park, FSID, IISc., under Prof. Chandan Kumar, CeNSE, IISc. in the quantum materials laboratory.
 - Worked on designing a Cryogenic Dip-Stick experimental setup to study the electronic properties of 2D materials at ultra-cold temperatures (using liq. N₂).
 - Done a theoretical study of the electronic transport properties of a few-layer (2D) NbSe₂ as an Ising Superconductor and its application as a Josephson junction and SQUID.
- **Naxxatra Sciences and Collaborative Research** March 2021 - December 2024
Research and Teaching Fellowship
Bengaluru, India.
 - *Project Organizer & Admin Role:* Led and managed scientific research initiatives as the Head of the Physics wing at Naxxatra Sciences, overseeing project development and collaboration.

- *Mentoring & Teaching*: Mentored students in online research projects and taught specialized courses in theoretical physics, such as 'Quantum Mechanics for Beginners', 'Astrophysics Summer Program', 'Astronomical Sources of Radio Waves', 'Stellar Evolution and Stellar Physics', 'Python for Scientists', 'General Relativity and its Applications', and 'Gravitational Waves & Black Holes'.
- *STEM Club Mentor*: Mentored the students of National Hill View Public School, Bengaluru, as a part of their STEM club. Guiding them to learn science scientifically and also through experiment-based learning.

SKILLS

- *Programming Languages*: Python, C++, C, Mathematica, Matlab, LaTeX, Git.

AWARDS

- **"Board Chairman's Gold Medal" & "K. Subbarayappa Gold Medal"** 23/11/2024
National Institute of Technology Karnataka Surathkal, India
Received "Board Chairman's Gold Medal" for scoring the highest CGPA 10/10 in the graduating batch of 2023-24 and the "K. Subbarayappa Gold Medal" for having the best academic record in M.Sc. Physics.
- **Gold Medals in Physics and Chemistry, and a Silver Medal in Mathematics** 26/04/2024
St. Joseph's College (currently St. Joseph's University) Bengaluru, India
Received Gold Medals in Physics & Chemistry, and a Silver Medal in Mathematics in the batch of 2019-22.
- **Secured 1st Place in Quiz Competition, World Asteroid Day 2021** 30/06/2021
REVA University, Bengaluru, India Bengaluru, India
- **Secured 1st Place in Presentation Competition, World Astroid Day 2021** 30/06/2021
REVA University, Bengaluru, India Bengaluru, India
- **Secured 2nd Place in online Tech Treasure Hunt, Tech Sparda 2022** 16/04/2022
National Institute of Technology Kurukshetra, Haryana, India Haryana, India
- **Got selected to the final round of International Physicists' Tournament 2022** February 2022
Captain of the Indian team that cleared the international pre-selection & got selected to the final international round.

CERTIFICATIONS

- **Tapestry of Field theory: Classical & Quantum, Equilibrium & Nonequilibrium Perspectives** January 2024 - April 2024
Marks: 76/100 (Elite + Silver)
Instructor: Prof. Mahendra Verma IIT Kanpur. Certifying Body: NPTEL Online Certification.

PERSONAL INFORMATION

- *Gender*: Male
- *Date of Birth*: September 05, 2000
- *Nationality*: Indian
- *Martial Status*: Single
- *Languages*: Kannada, English, Telugu, Hindi

REFERENCES

1. **Dr. Ajith K M** (*M.Sc. Thesis Supervisor*)
Professor, Department of Physics
National Institute of Karnataka, Surathkal, India
Email: ajith@nitk.ac.in
Phone: +91 9590885858
2. **Dr. V.SREENATH**
Assistant Professor, Department of Physics
National Institute of Karnataka, Surathkal, India
Email: sreenath@nitk.edu.in
Phone: +91 8242473276